

Introduction to Machine Learning

Homework 4

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Question 1

Let $X = \{x_1, \dots, x_N\}$ be the observed data. We introduce a latent binary variable $z_{nk} \in \{0, 1\}$ to indicate whether the data point x_n is generated by the k -th component, where $k = 1, \dots, K$ represents the Gaussian clusters and $k = K + 1$ represents the outlier component. For a given data point x_n , only one $z_{nk} = 1$ and the rest are 0. The prior probabilities are $p(z_{nk} = 1) = w_k$ with $\sum_{k=1}^{K+1} w_k = 1$. The conditional probabilities of the observations are:

$$p(x_n | z_{nk} = 1) = \begin{cases} \mathcal{N}(x_n; \mu_k, \Sigma_k) & \text{for } k = 1, \dots, K \\ p_{\text{noise}}(x_n) = \frac{1}{c^D} & \text{for } k = K + 1 \end{cases} \quad (1)$$

E-step: We calculate the responsibility $\gamma_{nk}^{(t)} = \mathbb{E}[z_{nk} | x_n, \theta^{(t)}]$, which is the posterior probability that x_n is generated by component k given the current parameters $\theta^{(t)}$. For the regular Gaussian components $k = 1, \dots, K$:

$$\gamma_{nk}^{(t)} = \frac{w_k^{(t)} \mathcal{N}(x_n; \mu_k^{(t)}, \Sigma_k^{(t)})}{\sum_{j=1}^K w_j^{(t)} \mathcal{N}(x_n; \mu_j^{(t)}, \Sigma_j^{(t)}) + w_{K+1}^{(t)} \frac{1}{c^D}} \quad (2)$$

For the outlier component $k = K + 1$:

$$\gamma_{n,K+1}^{(t)} = \frac{w_{K+1}^{(t)} \frac{1}{c^D}}{\sum_{j=1}^K w_j^{(t)} \mathcal{N}(x_n; \mu_j^{(t)}, \Sigma_j^{(t)}) + w_{K+1}^{(t)} \frac{1}{c^D}} \quad (3)$$

M-step: First, we write the complete data likelihood function $L(\theta)$:

$$L(\theta) = p(X, Z | \theta) = \prod_{n=1}^N \prod_{k=1}^{K+1} [w_k p(x_n | z_{nk} = 1)]^{z_{nk}} \quad (4)$$

Taking the logarithm, the complete data log-likelihood is:

$$\ln L(\theta) = \sum_{n=1}^N \sum_{k=1}^{K+1} z_{nk} \ln[w_k p(x_n | z_{nk} = 1)] \quad (5)$$

Then, we replace z_{nk} with its expected value $\gamma_{nk}^{(t)}$ conditionally on $\theta^{(t)}$ to obtain the expected complete data log-likelihood, making the function become the Q function:

$$Q(\theta, \theta^{(t)}) = \mathbb{E}_{Z|\theta^{(t)}}[\ln L(\theta)] = \sum_{n=1}^N \sum_{k=1}^{K+1} \gamma_{nk}^{(t)} \ln[w_k p(x_n | z_{nk} = 1)] \quad (6)$$

$$= \sum_{n=1}^N \sum_{k=1}^K \gamma_{nk}^{(t)} \ln[w_k \mathcal{N}(x_n; \mu_k, \Sigma_k)] + \sum_{n=1}^N \gamma_{n,K+1}^{(t)} \ln \left[w_{K+1} \frac{1}{c^D} \right] \quad (7)$$

Since $p_{\text{noise}}(x) = \frac{1}{c^D}$ is a constant with respect to the component parameters, it does not affect the derivative for μ_k and Σ_k . Let $N_k^{(t)} = \sum_{n=1}^N \gamma_{nk}^{(t)}$.

Updating the mixing weights w_k (using all $K+1$ components $k = 1, \dots, K+1$): By taking the derivative of Q with respect to w_k subject to the constraint $\sum_{k=1}^{K+1} w_k = 1$, we obtain the updated parameters at step $t+1$:

$$w_k^{(t+1)} = \frac{N_k^{(t)}}{N} = \frac{1}{N} \sum_{n=1}^N \gamma_{nk}^{(t)} \quad (8)$$

Updating μ_k and Σ_k (only for the first K components $k = 1, \dots, K$):

$$\mu_k^{(t+1)} = \frac{1}{N_k^{(t)}} \sum_{n=1}^N \gamma_{nk}^{(t)} x_n \quad (9)$$

$$\Sigma_k^{(t+1)} = \frac{1}{N_k^{(t)}} \sum_{n=1}^N \gamma_{nk}^{(t)} (x_n - \mu_k^{(t+1)})(x_n - \mu_k^{(t+1)})^T \quad (10)$$

Question 2

1. Mean-shift as a maximum likelihood estimation method

Let the probability density function be estimated by Kernel Density Estimation (KDE) with a Gaussian kernel: $\hat{p}(x) = \frac{1}{N} \sum_{n=1}^N K_h(x, x_n)$. To find the mode via Maximum Likelihood Estimation (MLE), we set $\nabla_x \hat{p}(x) = 0$. Using the derivation from Eq. (25), the gradient is:

$$\begin{aligned} \nabla_x \hat{p}(x) &= \frac{1}{N} \sum_{n=1}^N \nabla_x K_h(x, x_n) \\ &= -\frac{1}{Nh^2} \sum_{n=1}^N K_h(x, x_n)(x - x_n) \\ &\propto \sum_{n=1}^N K_h(x, x_n)(x_n - x) \\ &\propto \underbrace{\frac{\sum_{n=1}^N K_h(x, x_n)x_n}{\sum_{n=1}^N K_h(x, x_n)}}_{m(x)} - x = m(x) - x \end{aligned} \quad (11)$$

Setting this proportional gradient to 0 yields the fixed-point iteration step:

$$x^{(t+1)} = m(x^{(t)}) = \frac{\sum_{n=1}^N K_h(x^{(t)}, x_n)x_n}{\sum_{n=1}^N K_h(x^{(t)}, x_n)} \quad (12)$$

2. Modified mean-shift as MAP with prior $\mathcal{N}(\mu, \sigma^2)$

When x has a prior $p_{\text{prior}}(x) = \mathcal{N}(x; \mu, \sigma^2)$, we get $L(x) = \hat{p}(x)p_{\text{prior}}(x)$. So :

$$\begin{aligned}\nabla_x L(x) &= p_{\text{prior}}(x)\nabla_x \hat{p}(x) + \hat{p}(x)\nabla_x p_{\text{prior}}(x) \\ &= -\frac{1}{Nh^2}\mathcal{N}(\mathbf{x}) \sum_{n=1}^N K_h(x, x_n)(x - x_n) - \frac{(x - \mu)}{N\sigma^2}\mathcal{N}(\mathbf{x}) \sum_{n=1}^N K_h(x, x_n) \\ &\propto \frac{1}{h^2} \sum_{n=1}^N K_h(x, x_n)(x_n - x) + \frac{(\mu - x)}{\sigma^2} \sum_{n=1}^N K_h(x, x_n)\end{aligned}\tag{13}$$

Setting this proportional gradient to 0 yields the fixed-point iterative update rule:

$$x^{(t+1)} = \frac{\sigma^2 m(x^{(t)}) + h^2 \mu}{h^2 + \sigma^2}\tag{14}$$

3. Modified mean-shift as MAP with a GMM prior

Assume $p_{\text{prior}}(x) = \frac{1}{K} \sum_{k=1}^K \mathcal{N}_k(x)$, where $\mathcal{N}_k(x) = \mathcal{N}(x; \mu_k, \sigma_k^2)$. We denote the unified responsibility component as $f_k(x) = \frac{\mathcal{N}_k(x)}{\sum_{j=1}^K \mathcal{N}_j(x)}$. To maximize $L(x) = \hat{p}(x)p_{\text{prior}}(x)$, we first calculate the gradient of the prior:

$$\nabla_x p_{\text{prior}}(x) = \frac{1}{K} \sum_{k=1}^K \nabla_x \mathcal{N}_k(x) = p_{\text{prior}}(x) \sum_{k=1}^K f_k(x) \frac{\mu_k - x}{\sigma_k^2}\tag{15}$$

Substituting this into the posterior gradient $\nabla_x L(x)$, we merge the terms to extract the update target:

$$\begin{aligned}\nabla_x L(x) &= p_{\text{prior}}(x)\nabla_x \hat{p}(x) + \hat{p}(x)\nabla_x p_{\text{prior}}(x) \\ &= -\frac{1}{Nh^2}p_{\text{prior}}(x) \sum_{n=1}^N K_h(x, x_n)(x - x_n) + \hat{p}(x)p_{\text{prior}}(x) \sum_{k=1}^K f_k(x) \frac{\mu_k - x}{\sigma_k^2} \\ &\propto \frac{1}{h^2} \sum_{n=1}^N K_h(x, x_n)(x_n - x) + \left(\sum_{n=1}^N K_h(x, x_n) \right) \sum_{k=1}^K f_k(x) \frac{\mu_k - x}{\sigma_k^2} \\ &\propto \left(\frac{m(x)}{h^2} + \sum_{k=1}^K \frac{f_k(x)\mu_k}{\sigma_k^2} \right) - \left(\frac{1}{h^2} + \sum_{k=1}^K \frac{f_k(x)}{\sigma_k^2} \right) x \\ &\propto \frac{\frac{m(x)}{h^2} + \sum_{k=1}^K \frac{f_k(x)\mu_k}{\sigma_k^2}}{\frac{1}{h^2} + \sum_{k=1}^K \frac{f_k(x)}{\sigma_k^2}} - x\end{aligned}\tag{16}$$

By setting this proportional term to 0, we obtain the fixed-point iteration rule:

$$x^{(t+1)} = \frac{\frac{m(x^{(t)})}{h^2} + \sum_{k=1}^K \frac{f_k(x^{(t)})}{\sigma_k^2} \mu_k}{\frac{1}{h^2} + \sum_{k=1}^K \frac{f_k(x^{(t)})}{\sigma_k^2}}\tag{17}$$